Training Leaves Traces: White-Box Model Lineage Verification using Diagonal-Dominant Fingerprints

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Abstract

Open-weight checkpoints are routinely fine-tuned, quantized, pruned, merged, and redistributed without reliable lineage metadata. Existing provenance tools are brittle: hashes fail after any weight change, metadata requires trust, and watermarks must be inserted before release. **We study a post-hoc white-box weight-lineage verification: given two checkpoints with compatible residual architecture, do they share weight-level lineage after common checkpoint transformations?** This is distinct from behavioral similarity: a distilled student imitates its teacher's outputs, but its weights are trained from scratch and share no lineage.

We show that residual-network training induces an intrinsic fingerprint: correctly paired residual-branch products develop a diagonal-dominant fingerprint, measured by a trace/Frobenius score, that is absent at initialization and in non-residual controls. We use this signal to align blocks between checkpoints and compute a model-level lineage score from centered branch signatures. Across 12 architectures spanning language, vision, and speech (GPT-2 to LLaMA-2, ViT, Whisper, ResNets), the fingerprint recovers correct projection pairings with 91–100% accuracy, and the lineage score achieves complete separation: fine-tuned, pruned, quantized, and LoRA-merged checkpoints score high, while independently trained models and distilled students remain low. Unlike activation-based CKA and IPGuard, which track functional similarity, this provides a passive, retroactive provenance signal for white-box residual checkpoints.

1 Introduction

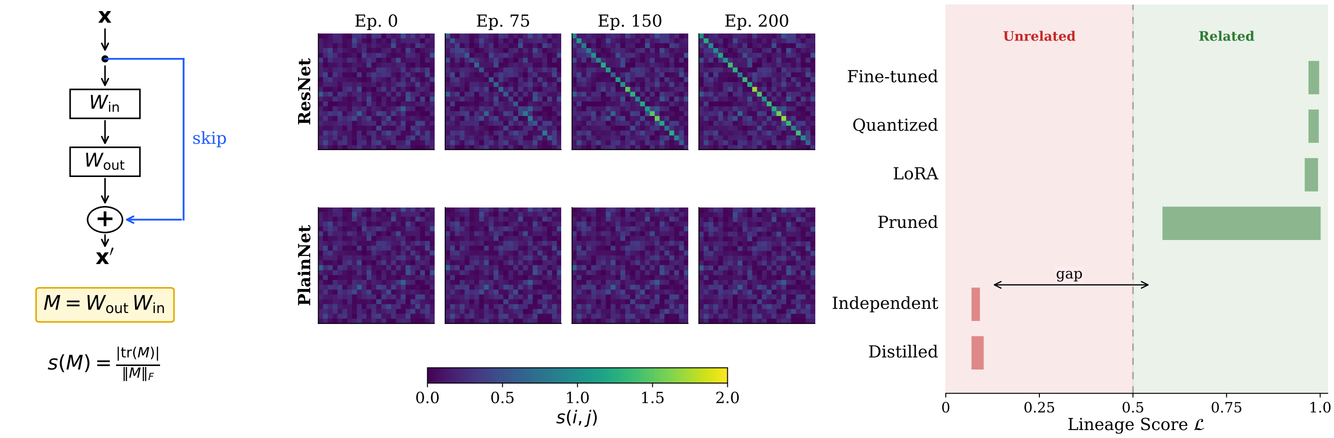
The open-weight ecosystem has transformed how neural networks are distributed and reused. When Meta released LLaMA (Touvron et al. 2023), the community produced instruction-tuned variants within weeks. Platforms like Hugging Face now host hundreds of thousands of model checkpoints (Hugging Face 2024), many of which are fine-tuned (Devlin et al. 2019; Ouyang et al. 2022), quantized (Han, Mao, and Dally 2016; Dettmers et al. 2022), pruned (Han et al. 2015), or merged (Wortsman et al. 2022; Ilharco et al. 2023). This creates a *model supply chain* where downstream checkpoints inherit weights from upstream models, but the lineage is often undocumented or intentionally obscured. Provenance of AI-generated content is now treated as a first-class problem: Google's SynthID watermarks text and images at generation time (Dathathri et al. 2024), yet no analogous mechanism exists for the models themselves. Model reuse is easy; reliable model provenance is not.

Existing provenance mechanisms fall short (Table 1). Cryptographic hashes break after any weight modification. Metadata and model cards (Mitchell et al. 2019) require honest distributors and preserved records; they can be falsified or lost. Watermarks (Uchida et al. 2017; Adi et al. 2018) must be inserted before release; they cannot be applied retroactively to already-deployed checkpoints, and no surveyed scheme proved robust against fine-tuning or pruning (Lukas et al. 2022). Behavioral tests that compare model outputs cannot distinguish weight inheritance from imitation: two models can behave similarly without sharing weights, especially after distillation (Hinton, Vinyals, and Dean 2015). Activation-based similarity measures like CKA (Kornblith et al. 2019) and decision-boundary fingerprints like IPGuard (Cao, Jia, and Gong 2021) inherit this limitation. We need a passive, post-hoc signal that reads provenance from the weights themselves.

We ask: given a reference checkpoint and a suspect checkpoint with the same residual architecture (e.g., both LLaMA-2-7B derivatives), can we determine whether the suspect inherited weights from the reference, even after common transformations like fine-tuning, quantization, or pruning? This is distinct from duplicate detection (which hashes solve until the first modification), behavioral similarity (which conflates imitation with inheritance), and watermark verification (which requires forethought). The question is whether shared weight ancestry leaves detectable structure after modification.

The answer is yes. We find that training itself leaves a characteristic fingerprint in the weights of residual networks. In these architectures (He et al. 2016), each block adds a learned transformation to its input via a skip connection (Figure 1a). During training, the input and output projections of each block align so that their product $W_{\text{out}} W_{\text{in}}$ develops diagonal-dominant structure that is absent at random initialization and does not appear in networks without skip connections (Figure 1b). This structure emerges from the optimization process: it is learned, not an initialization artifact (Section 3).

We exploit this fingerprint for provenance verification. Each residual block yields a signature: its branch product



(a) Residual Branch Product **(b) Training Induces Diagonal Dominance: ResNet vs PlainNet** **(c) White-Box Model Lineage Verification**

Figure 1: Training leaves traces in residual networks. (a) The residual branch product $M = W_{\text{out}}W_{\text{in}}$ for a block with skip connection; the diagonal-dominance score $s(M)$ measures trace concentration. (b) Training dynamics: score matrices $s(i, j)$ from epoch 0 to 5. ResNet (top row) develops strong diagonal structure (100% block-pairing accuracy); PlainNet without skip connections (bottom row) remains random (3% accuracy). The fingerprint is learned, not an initialization artifact. (c) Lineage verification: related checkpoints (fine-tuned, quantized, pruned, LoRA-merged) score high on $\mathcal{L}$; unrelated checkpoints (independent, distilled) score low, enabling clean separation above/below threshold.

scored for diagonal concentration via a trace/Frobenius ratio (Equation 3). Correctly paired blocks from the same trained model score high, while mismatched or unrelated blocks do not. The extraction is architecture-aware, respecting the specific structure of Transformer MLPs, attention paths, and ResNet bottleneck blocks (Table 2). To compare two checkpoints, we align their blocks using these signatures, aggregate the pairwise similarities into a model-level lineage score, and threshold it against a null distribution built from unrelated models (Equation 7).

The fingerprint is not specific to one model family. We observe it in public checkpoints across language, vision, and speech (GPT-2, BERT, LLaMA-2, Mistral, Qwen2.5, DeepSeek-R1, ViT, Whisper, and ResNets) where it recovers block correspondences with over 91% accuracy (Table 3). Turning to the verification task itself, we construct controlled lineage benchmarks on MLPs and ResNet-18: models with genuine weight ancestry (produced by fine-tuning, weight noise, pruning, quantization, or LoRA merging) are cleanly separated from those without (independent training runs and distilled students), with AUROC = 1.000. The distillation case is instructive: a student can closely replicate its teacher's outputs, yet our score correctly labels it unrelated, because its weights carry no trace of the teacher's training (Figure 1c). Finally, on the public LLaMA-2 family, the method flags three real fine-tuned descendants (chat, Vicuna, CodeLlama) as related while rejecting four independently trained checkpoints, among them OpenLLaMA, which shares the architecture but inherits no weights.

Our contributions:

- **Diagonal-dominant fingerprint.** We establish that gradient-based training drives the input and output projections of residual blocks toward mutual alignment, concentrating their product along the diagonal, a property absent at random initialization and never acquired by skip-free architectures.
- **White-box verification method.** We convert this signature into a practical test: per-block scores are matched across two checkpoints, aggregated into a single lineage statistic, and judged against a null distribution, requiring no watermark, no metadata, and no cooperation from the model's distributor.
- **Broad empirical validation.** We evaluate on 280+ checkpoint pairs covering 12 architectures and 5 model families, from synthetic benchmarks with known ground truth to real-world derivatives of LLaMA-2, obtaining 91–100% block correspondence and AUROC=1.000.
- **Robustness and specificity.** The fingerprint persists through fine-tuning, quantization, pruning, and LoRA merging, yet vanishes for distilled students and independently trained models of identical architecture, a distinction beyond the reach of behavioral measures such as CKA and IPGuard.

2 Problem Setting and Related Work

This section formalizes the lineage verification problem and positions it against prior provenance mechanisms.

2.1 White-Box Model Lineage Verification

Scope. A verifier is given two checkpoints A and B and must determine whether they share weight-level lineage. We assume white-box access to both weight sets and compatible residual architectures (same layer count L and hidden dimension d); cross-architecture comparisons are out of scope. Metadata may be missing or adversarial. The verifier operates on weights alone: no training data, activations, or forward passes are available.

Method	Retro.	FT	Quant	Distill
Crypto Hash	✓	✗	✗	✓
Metadata/Logs	✓	✓	✓	✓
Watermarking	✗	∼	∼	✗
Decision-Boundary	✓	✓	✓	✗
Ours	✓	✓	✓	✓

Table 1: Comparison with prior provenance methods. Retro.: applicable retroactively. FT: survives fine-tuning. Quant: survives quantization. Distill: correctly classifies distilled copies as unrelated. $\sim$: partial or unreliable.

Architecture	Residual branch $F(x)$	Product M
Transformer MLP / BasicBlock	$W_2\sigma(W_1x)$	W_2W_1
Attention V/O path	$W_O\mathrm{Attn}(x)W_Vx$	W_OW_V
Attention Q/K path	bilinear $x^\top W_QW_K^\top x$	$W_QW_K^\top$
Bottleneck ResNet	$W_3\sigma(W_2\sigma(W_1x))$	$W_3W_2W_1$
SwiGLU MLP	$W_{\mathrm{down}}[\sigma(W_{\mathrm{gate}}x)\odot W_{\mathrm{up}}x]$	$W_{\mathrm{down}}W_{\mathrm{up}}$

Table 2: Architecture-aware residual branch products. Each M composes the linear maps of one branch (dropping nonlinearities) so it maps the residual stream to itself. Factorization must match the architecture; incomplete products destroy the signal (Section 5).

Relatedness. Checkpoints are *related* if they share a common weight ancestor: either one derives from the other, or both derive from a shared parent, through a chain of transformations $\theta' = T(\theta, D)$, where T may include continued optimization (fine-tuning, RLHF), compression (pruning, quantization), or adapter merging (LoRA), and D is optional additional data. Checkpoints are *unrelated* if their weights were initialized independently, even when they share the same architecture, training data, task, or output behavior. Transformations with multiple parents, such as linear merges of two checkpoints, induce partial lineage between the suspect and each parent; we quantify this regime in Appendix D.5.

The verifier solves lineage verification: do A and B share weight ancestry? It returns one of three verdicts: RELATED, UNRELATED, or INCONCLUSIVE (insufficient evidence either way). The lineage score is symmetric, $\mathcal{L}(A, B) = \mathcal{L}(B, A)$: the protocol establishes whether two checkpoints share ancestry, not which is the ancestor; directional attribution requires metadata or additional candidates (Appendix C). As a by-product, the verification pipeline recovers block-level correspondences between the two checkpoints (Section 4.1), which serve as a diagnostic when the verdict is RELATED.

Lineage follows initialization, not behavior. Under this definition, what determines lineage is where the weights came from, not what the model does. A distilled student trained from freshly initialized weights is unrelated to its teacher, no matter how closely it imitates the teacher's outputs; model-extraction copies that replicate a model's function through query access (Tramèr et al. 2016) are likewise unrelated. Conversely, a student initialized from its teacher's weights is weight-related regardless of the distillation objective: initialization, not training procedure, determines lineage. The method is thus a forensic primitive for weight-level provenance, complementing cryptographic signing, metadata, and watermarking where those were never applied; it is not a legal ownership proof.

2.2 Related Work

We organize prior work by the signal each method measures; Table 1 summarizes the comparison.

Cryptographic and metadata provenance. Cryptographic hashes verify exact byte-level identity but fail after any weight modification: the avalanche property ensures that fine-tuning, pruning, or even floating-point rounding produces an entirely different digest. Model cards (Mitchell et al. 2019) and experiment-tracking systems document provenance but require honest distributors and preserved records; they can be falsified or lost when checkpoints are redistributed outside controlled environments. Proof-of-learning protocols (Jia et al. 2021) verify that a claimed party performed the training, but require retained training transcripts: forethought that already-released checkpoints lack.

Embedded ownership signals. Parameter watermarks (Uchida et al. 2017) and backdoor watermarks (Adi et al. 2018) embed deliberate signals during training for later extraction. Lukas et al. (2022) found no surveyed scheme reliably robust against fine-tuning and pruning within realistic compute budgets, and all require proactive insertion: unwatermarked legacy checkpoints cannot be verified retroactively. Output watermarks like SynthID mark AI-generated content, not weights: a different problem.

Behavioral and decision-boundary fingerprints. IPGuard (Cao, Jia, and Gong 2021) extracts inputs near the classification boundary as fingerprints; adversarial frontier stitching (Le Merrer, Perez, and Trédan 2020) uses adversarial examples to mark decision surfaces. These measure functional similarity and cannot distinguish weight inheritance from imitation: a distilled student may share decision surfaces while having entirely independent weights.

Representation similarity. CKA (Kornblith et al. 2019) and SVCCA (Raghu et al. 2017) compare learned representations using activation patterns on a probe dataset. They may regard a successful distilled student as similar: reasonable for their objective (representation alignment) but not ours (weight ancestry). They also require data and forward passes, whereas our method operates on weights alone.

Weight-space provenance. Generic parameter-space similarity (weight cosine, Frobenius distance, permutation-aligned matching (Ainsworth, Hayase, and Srinivasa 2023)) can detect derived checkpoints when the suspect remains close to the reference in parameter space. But these are uncalibrated global scalars: they localize nothing, offer no account of why proximity implies lineage or when it stops doing so, and their margins are not grounded in any property of training. A recent line builds purpose-built provenance on weights, but each requires something beyond the two checkpoints: HuRef (Zeng et al. 2025) relies on convergence of LLM parameter directions after pretraining; REEF (Zhang et al. 2024) and neural lineage detection (Yu and Wang 2024) need probe data, activations, or gradients; MoTHer (Horwitz, Shul, and Hoshen 2025) recovers directed model trees but requires a collection of checkpoints rather than a single pair. None of these can say which weights were inherited. Our fingerprint is read block by block from trained residual structure: it is data-free, compares a single pair of checkpoints against a calibrated null, and reveals not only whether a suspect inherited weights but which blocks it inherited and how many.

3 Training-Induced Diagonal Dominance in Residual Branch Products

A residual block computes $x_{\ell+1} = x_\ell + F_\ell(x_\ell)$, where F_ℓ is the residual branch. For a branch with K linear layers $W_1, \ldots, W_K$ interleaved with nonlinearities, we define the *residual branch product* by composing the linear factors and dropping the nonlinearities between them:

$$M_\ell = W_K W_{K-1} \cdots W_1 \in \mathbb{R}^{d\times d} \tag{1}$$

Individual factors W_k are rectangular and do not share an input–output basis; their composed product maps the residual stream back into its own coordinate space, so its diagonal entries compare like with like. The exact factorization is architecture-dependent (Table 2).

3.1 Why Residual Training Induces Diagonal Dominance

Under the Frobenius inner product $\langle A, B\rangle = \mathrm{tr}(A^\top B)$, the identity matrix and the traceless matrices span orthogonal subspaces of $\mathbb{R}^{d\times d}$. Any branch product therefore splits into a component along the identity, with coefficient $\langle M_\ell, I\rangle / \langle I, I\rangle = \mathrm{tr}(M_\ell)/d$, and an orthogonal traceless remainder:

$$M_\ell = -\varepsilon_\ell I + E_\ell \tag{2}$$

where $\varepsilon_\ell = -\mathrm{tr}(M_\ell)/d$ and $\mathrm{tr}(E_\ell) = 0$. The sign convention absorbs the empirical regularity that traces of trained branches are predominantly negative (Section 5), making ε_ℓ typically positive; nothing downstream depends on the sign, since the score below uses the trace magnitude. The decomposition itself is a coordinate choice, available for any matrix.

We measure this concentration with the *diagonal-dominance score*

$$s(M) = \frac{|\mathrm{tr}(M)|}{\|M\|_F}, \tag{3}$$

the fraction of the matrix's energy lying along the identity direction, up to a factor of $\sqrt{d}$. The finding is that training moves energy into the identity component: in trained residual networks, correctly paired branch products score far above unpaired or untrained ones, a concentration absent at initialization and never developed without skip connections (Figure 1b).

The skip connection explains why. It places the learned branch in an identity-centered parameterization: training optimizes a correction around an explicit identity path rather than learning the full transformation from scratch. For stable gradient flow, the block Jacobian $J = I + J_F$ should remain near-orthogonal throughout training, a condition known as dynamical isometry (Pennington, Schoenholz, and Ganguli 2017). If $J \approx I$, then $J_F \approx 0$; for small but nonzero learned corrections, the linear component of J_F tends toward $-\varepsilon I$ with small ε. Because W_{in} and W_{out} receive coupled gradients through the branch, this pressure aligns their product with coordinate-preserving directions. Non-residual networks lack this constraint: without the skip connection, there is no functional pressure toward identity, and nothing couples W_{in} with W_{out} more than with any other weight matrix.

This is a mechanistic hypothesis, not a theorem, but each of its predictions is testable. It predicts no signal at initialization under any scheme: an ablation over seven initializations finds chance-level pairing before training, including orthogonal initialization, which satisfies dynamical isometry by construction yet has learned nothing (Saxe, McClelland, and Ganguli 2014) (Appendix A.1). It predicts no signal without the skip connection: a PlainNet trained to comparable loss develops none (Appendix A.2).

3.2 Margin Analysis

For the score to separate correct pairings from incorrect ones, two things must hold: correctly paired products must score high, and mismatched products must score low. We treat each in turn.

By the orthogonality of decomposition (2), $\|M_\ell\|_F^2 = \varepsilon_\ell^2 d + \|E_\ell\|_F^2$: the branch product's energy splits exactly between its identity component and its traceless remainder. The score therefore depends on M only through the ratio of the two:

Proposition 1 (Correct-pair score)**.** *Let $M = -\varepsilon I + E$ with $\mathrm{tr}(E) = 0$ and $\varepsilon \neq 0$. Then:*

$$s(M) = \frac{\sqrt{d}}{\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}} \leq \sqrt{d} \tag{4}$$

with equality if and only if $E = 0$.

Proof. $|\mathrm{tr}(M)| = |\varepsilon| d$ and $\|M\|_F^2 = \varepsilon^2 d + \|E\|_F^2$ by orthogonality. Substitute into (3).

The score is $\sqrt{d}$ discounted by how much the learned residue E outweighs the identity component: geometrically, $s(M)/\sqrt{d}$ is the cosine of the angle between M and the identity. A correct pair scores high exactly to the extent that training concentrated its product along the diagonal.

A mismatched product has no such concentration. For W_{out} and W_{in} drawn from different blocks, the product has no preferred alignment with the identity: its trace is a sum of d weakly correlated terms of comparable size to the other d^2 entries, giving

Corollary 1 (Mismatched baseline)**.** *For products of projections from independent blocks, $\mathbb{E}[s] \approx 1/\sqrt{d}$.*

The margin between the two regimes therefore scales as d: wider networks separate correct from incorrect pairings more cleanly, not less. Empirically the correct-pair score grows as $d^{0.87}$ (Section 5), faster than the $\sqrt{d}$ envelope's rate, consistent with $\|E\|_F/(\varepsilon\sqrt{d})$ shrinking as d grows, while always remaining below the $\sqrt{d}$ bound. The proposition bounds the score; it does not fix its growth rate.

4 From Diagonal-Dominance Fingerprints to Lineage Verification

Diagonal dominance identifies correctly paired trained residual branches, but it does not establish lineage: unrelated trained models both exhibit it. The decomposition (2) supplies the division of labor. The identity component is generic; its strength certifies that a branch carries trained residual structure, and Section 5.1 uses it to recover projection pairings within a model. But precisely because it is shared across trained models, it says nothing about ancestry. The traceless component E_ℓ is block- and checkpoint-specific: it survives weight-preserving transformations but not reinitialization. Verification therefore runs on the traceless component, with diagonal dominance serving as a gate on which branches are trustworthy. This section builds the protocol in three steps: signatures, alignment, decision.

4.1 Centered Residual Signatures and Block Alignment

For each branch product M_ℓ we retain only the traceless remainder of decomposition (2), normalized to a unit vector:

$$R_\ell = M_\ell - \frac{\operatorname{tr}(M_\ell)}{d} \cdot I, \quad \phi_\ell = \frac{\operatorname{vec}(R_\ell)}{\|\operatorname{vec}(R_\ell)\|_2} \tag{5}$$

R_ℓ is exactly the E_ℓ of (2); the signature ϕ_ℓ carries the block- and checkpoint-specific geometry that weight-preserving transformations inherit and independent training cannot reproduce. Because M_ℓ is exactly invariant under the two function-preserving reparameterizations of a residual branch (hidden permutation: $W_{\text{in}} \leftarrow PW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}P^\top$; reciprocal rescaling: $W_{\text{in}} \leftarrow cW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}/c$), so is every signature built on it: re-basing hidden units cannot evade the comparison (Section 6).

Given reference A and suspect B, each with L blocks, we form the signature-similarity matrix $G_{ij} = \langle \phi_i^A, \phi_j^B \rangle$, gated so that branches with weak diagonal dominance in either checkpoint contribute conservatively (Appendix C), and recover the block correspondence as the assignment maximizing total similarity,

$$\pi^* = \arg\max_{\pi \in \mathcal{P}_L} \sum_i G_{i,\pi(i)}, \tag{6}$$

via the Hungarian algorithm (Kuhn 1955). For derived checkpoints that preserve block ordering, π^* should be the identity permutation, and recovering it without metadata is evidence that the signatures track block identity rather than position; the Hungarian step matters when ordering cannot be trusted, and its output doubles as a diagnostic: when the verdict is RELATED, the correspondence matrix shows which blocks were inherited (Appendix A). Computing the branch products costs $O(Ld^2h)$, where h is the branch's inner width; the similarity matrix adds $O(L^2d^2)$ and the assignment a negligible $O(L^3)$.

4.2 Lineage Score and Verification Test

The lineage score averages signature similarity over aligned branches:

$$\mathcal{L}(A, B) = \frac{1}{L} \sum_{\ell=1}^{L} \langle \phi_\ell^A, \phi_{\pi^*(\ell)}^B \rangle \tag{7}$$

Each term is a cosine, so $\mathcal{L} \in [-1, 1]$ with $\mathcal{L}(A, A) = 1$. The score is symmetric, $\mathcal{L}(A, B) = \mathcal{L}(B, A)$: exchanging the roles of A and B transposes G, and the optimal assignment value is invariant under transposition, provided the gate is defined over both checkpoints jointly (Appendix C). Related checkpoints should score near 1 and unrelated ones near 0, since independently initialized weights produce uncorrelated E_ℓ; Section 5 tests both predictions.

To map the score onto the verdicts of Section 2, we calibrate against a null distribution: with $\mathcal{N}_A = \{\mathcal{L}(A, U_j)\}$ the scores between A and independently trained models U_j, and (μ_0, σ_0) its mean and standard deviation,

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_0}{\sigma_0} \tag{8}$$

The protocol returns

$$V(A, B) = \begin{cases} \textsc{Related} & \text{if } Z > \tau_{\text{upper}} \\ \textsc{Unrelated} & \text{if } Z < \tau_{\text{lower}} \\ \textsc{Inconclusive} & \text{otherwise} \end{cases} \tag{9}$$

with symmetric thresholds $\tau_{\text{upper}} = 3.0$, $\tau_{\text{lower}} = -3.0$, giving a false-positive rate below 0.13% under a Gaussian null. Compatibility (same layer count L and hidden dimension d, per the setting of Section 2) is a precondition checked before any scoring; the protocol returns INCOMPATIBLE when it fails, an enforcement of the setting rather than a fourth verdict. The verifier needs weights alone, and $O(L^2 + d^2)$ working memory beyond the checkpoints themselves.

5 Experiments

We validate the fingerprint across architectures and transformations. Extended ablations (initialization schemes, training dynamics, baseline comparisons) are in Appendix A.

5.1 Cross-Architecture Generalization

Table 3 reports block-pairing accuracy across 12 architectures spanning language (GPT-2, BERT, LLaMA-2, Mistral, Qwen, DeepSeek), vision (ViT, ConvNeXt, ResNet), and speech (Whisper). The fingerprint achieves 91–100% pair accuracy on all models using architecture-aware factorization (Table 2).

Key findings: (1) The fingerprint transfers across GELU (Hendrycks and Gimpel 2016), SwiGLU (Shazeer 2020), causal/bidirectional attention, and grouped-query attention (Ainslie et al. 2023). (2) RLHF preserves the signal (LLaMA-2 base→chat: 100% accuracy), while a Llama-3-derived DeepSeek distill is correctly classified as unrelated, isolating weight inheritance from behavioral pedigree. (3) For ResNet Bottleneck blocks, the naive W_3W_1

Model	Layers	Pair Acc	AUC
GPT-2 (124M–1.5B)	12–48	**100%**	1.00
BERT-base / ViT-B/16	12	**100%**	0.97–1.00
Mistral-7B / LLaMA-2-7B	32	**100%**	0.96–1.00
LLaMA-2-7B-chat (+RLHF)	32	**100%**	0.96–1.00
Qwen2.5-7B / DeepSeek-R1	28–32	80–**100%**	0.93–1.00
Whisper (tiny/base/small)	4–12	**100%**	0.85–1.00
ResNet-50/101/152	5–35	91–100%	0.96–1.00

Table 3: Cross-architecture generalization. Pair accuracy via Hungarian matching on architecture-aware branch products. Sub-100% SwiGLU sub-paths are rescued by joint factorization. Random-init baselines: $\leq 9\%$.

Transformation	Mean $\mathcal{L}$	Min $\mathcal{L}$	Det. @ 1%FPR
Fine-tune / Quant / Noise	0.99	0.97	45/45
Fine-tune (diff. target)	0.94	0.84	15/15
Pruning (10–85%)	0.81	0.58	15/15
Independent / Distilled	0.08	0.01	-

Table 4: Lineage score survival. All 75 related checkpoints detected at 1% FPR; unrelated checkpoints stay below 0.20.

fails; the correct $W_3W_2W_1$ recovers 91–100%. (4) Mean correct-pair $s(i,i)$ grows from 4.18 at d=768 (GPT-2) to 23.83 at d=5120 (LLaMA-2-13B), an empirical $d^{0.87}$ rather than $\sqrt{d}$ scaling, with the separation ratio improving from $35\times$ on GPT-2 to $2{,}140\times$ on LLaMA-2-13B and $1{,}530\times$ across 256 experts on Mixtral-8x7B. (5) As a real-LLM lineage check at the 7B scale, the residual-signature score $\mathcal{L}$ correctly identifies three *real fine-tuned* variants of LLaMA-2-7B—the RLHF chat model ($\mathcal{L} = 0.995$), the Vicuna instruction fine-tune ($\mathcal{L} = 0.996$), and CodeLlama ($\mathcal{L} = 0.336$)—as RELATED, while four independently-trained negatives (Mistral-7B, OpenLLaMA-7B, Yi-6B, and a Llama-3-derived DeepSeek distill) all score $\mathcal{L} \approx 4 \times 10^{-4}$—UNRELATED against the real cross-model null they define. OpenLLaMA, a from-scratch model with LLaMA-2's exact architecture, is the decisive negative: it is classified as unrelated on weights alone, since its configuration is indistinguishable from a related checkpoint. Details in App. D.4.

5.2 Robustness Under Post-Training Modification

We test survival under five transformation families on depth-24 MLPs: fine-tuning (same/different target), Gaussian noise ($\sigma_{\text{rel}} \leq 15\%$), magnitude pruning (10–85%), and quantization (16–256 levels).

Table 4 shows 100% detection at 1% FPR across all transformations. The weakest related checkpoint (85% pruning, $\mathcal{L}$=0.58) remains $7\times$ above the strongest unrelated checkpoint (distilled, $\mathcal{L}$=0.19). Fingerprint degradation correlates with utility loss (ρ=-0.83): suppressing the signal damages the model.

Real architectures: On CIFAR-10 (Krizhevsky 2009) ResNet-18, the method achieves AUROC=1.000 with

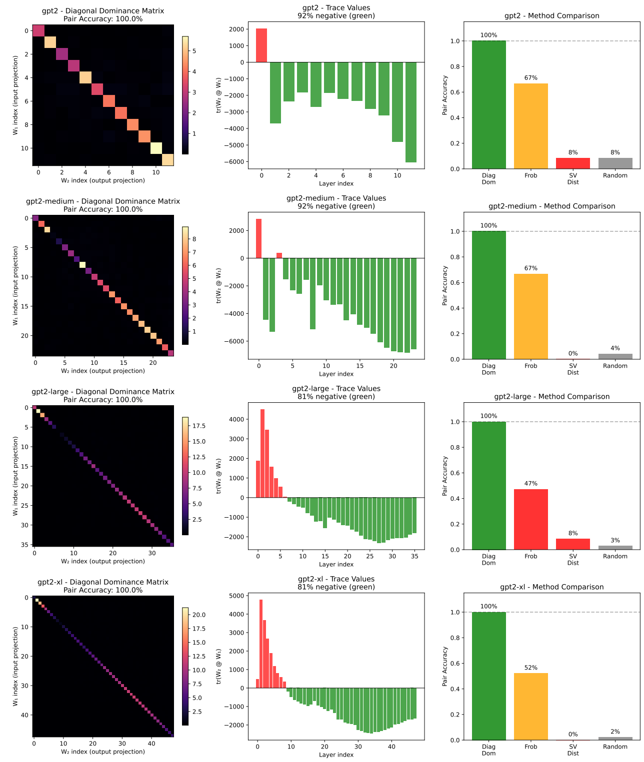


Figure 2: GPT-2 pairing. Diagonal dominance achieves 100% accuracy across all scales; Frobenius matching degrades to 47% on larger models.

$>$100$\times$ separation between related and unrelated checkpoints (Appendix A).

5.3 Comparison with Modern Baselines

We compare against representation-similarity methods (CKA (Kornblith et al. 2019), SVCCA (Raghu et al. 2017)), weight-space baselines (aligned Frobenius via Hungarian alignment, singular-value distance, weight cosine), and a decision-boundary baseline (IPGuard (Cao, Jia, and Gong 2021), adapted for regression via quantile-bin agreement) on a 52-pair MLP lineage benchmark (five related-checkpoint transform kinds plus different-seed and distilled negatives; per-kind composition in App. D.3) (Table 5). Four data-free weight-space methods-including ours, weight cosine, and aligned Frobenius-reach AUROC=1.000 on this benchmark, so the headline number cannot itself distinguish them. The standardized Gap-Z column makes the fragility of the non-weight-space methods explicit: IPGuard ranks truly related checkpoints *below* distilled unrelated ones (Gap-$Z = -3.5$), reproducing the structural distillation failure of decision-boundary methods; CKA's Gap-$Z = +8.6$ is large in standard-deviation units only because $\sigma_{\text{distill}} = 0.014$ is tiny, while the underlying raw ratio is $0.999/0.827 \approx 1.21\times$, so any benchmark reweighted toward distillation would push CKA's AUROC well below 1.000. We do not claim that diagonal dominance beats weight cosine on aggregate AUROC; on a benchmark where related checkpoints

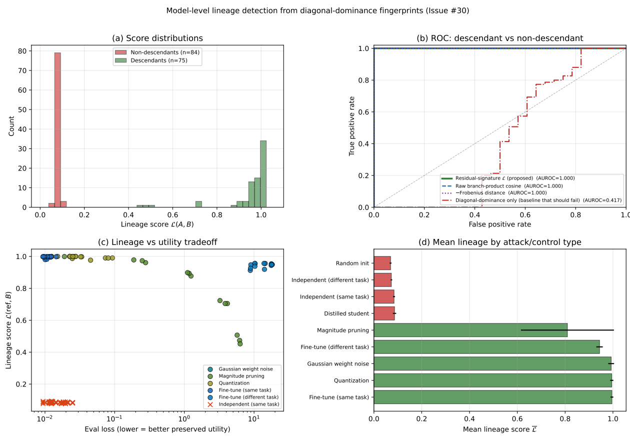


Figure 3: Lineage detection on depth-24 residual MLPs. (a) Score distributions: related (green, n=75) and unrelated (red, n=84, including independent and distilled) separate completely. (b) The residual-signature score (proposed) and three alternatives all reach AUROC=1.000, except the diagonal-dominance-only baseline (AUROC=0.417) which cannot distinguish two trained residual models. (c) Lineage vs. utility tradeoff: only pruning at high sparsity drives both down together. (d) Mean lineage by control type: related ≥ 0.81, unrelated ≤ 0.09.

are mild perturbations of the reference, a flattened cosine is a strong baseline. The contribution is rather that the residual-product signature (i) yields per-block correspondence as a by-product (only aligned Frobenius shares this; weight cosine and the activation methods do not), (ii) admits the closed-form margin analysis of Proposition 1, and (iii) is competitive with the strongest data-free weight-space baselines while remaining interpretable in terms of trained residual geometry. The harder synthetic regime of Appendix D.5 (layer grafts and linear merges) reinforces this positioning: three of the four data-free weight-space methods including ours (diagonal dominance, aligned Frobenius, weight cosine) remain at or near AUROC=1.000, while singular-value distance fails the merge regime (0.448) because spectral statistics are not linear in the weight-blend mixture, CKA drops to ≈ 0.80, and IPGuard collapses (AUROC 0.05 on grafts). The diagonal-dominance score additionally retains a near-perfect Spearman correlation ($\rho \approx 0.96$) with the fraction of reference blocks present in the suspect, supporting the interpretability claim that it tracks partial overlap monotonically rather than merely clearing a threshold.

5.4 Public Model Validation: LLaMA-2 Family

To move beyond controlled benchmarks, we evaluate lineage detection on the public LLaMA-2 family against the weight-space and decision-boundary baselines of Section 5.3 (Appendix D.4). Using LLaMA-2-7B as reference, we score three real fine-tuned variants—the RLHF chat model ($\mathcal{L} = 0.995$), the Vicuna-7B instruction fine-tune ($\mathcal{L} = 0.996$), and CodeLlama-7B ($\mathcal{L} = 0.336$), a code-specialized continued-pretraining derivative that is the weakest related checkpoint but still $\sim$700$\times$ above the null—against four independently-trained negatives: Mistral-7B, Yi-6B, DeepSeek-R1-Distill-8B, and, critically, OpenLLaMA-7B, a from-scratch reproduction of the LLaMA architecture that shares its exact d=4096, L=32 shape but inherits no weights. All four negatives score $\mathcal{L} \in [3.5, 4.8] \times 10^{-4}$, defining a real cross-model null ($\mu_0 = 4.2 \times 10^{-4}$); with only four real independents this is a descriptive separation check, not a calibrated false-positive rate. Diagonal dominance separates related from unrelated checkpoints by $\approx$1850$\times$ in mean score, versus 6$\times$ for weight cosine on all seven pairs and 2$\times$ for IPGuard on the logit subset where it is defined; aligned Frobenius and singular-value distance fail to separate this real-model set (AUROC=0.667), and the activation baselines (CKA/SVCCA) are omitted at 7B for cost. The OpenLLaMA case is decisive: identical architecture, zero inherited weights, correctly classified as unrelated—the verdict comes from the weights, not the configuration. This provides descriptive evidence that the margin advantage observed on controlled benchmarks transfers to real 7B-scale foundation models. Details in Appendix D.4.

6 Robustness and Boundary Cases

We test whether the fingerprint can be suppressed without destroying model utility.

Function-preserving symmetries. Per-block permutations ($W_{\text{in}} \leftarrow PW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}P^{\top}$) and cross-layer rescaling ($W_{\text{in}} \leftarrow cW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}/c$) leave the branch product $M = W_{\text{out}}W_{\text{in}}$ exactly unchanged, so $\mathcal{L} = 1.0$ under these attacks. Orthogonal rotations also preserve M but break model function.

Gradient-based suppression. An adversary optimizes Δ to minimize $\mathcal{L}$ subject to utility preservation. Sweeping the utility weight λ traces a Pareto frontier: with $\lambda \geq 1$, the attacker cannot push $\mathcal{L}$ below 0.91; suppressing to chance level ($\mathcal{L} \approx 0.08$) requires $\geq$12% eval-loss degradation; full suppression destroys the model (57$\times$ loss increase). Details in Appendix E.

Distillation boundary. The fingerprint tracks weight inheritance, not functional imitation. Distilled students ($\mathcal{L} = 0.086$) are correctly classified as unrelated in our same-architecture controlled benchmark—they trained fresh weights and share no residual structure with the teacher, even though they replicate its outputs. Cross-architecture distillation (e.g., BERT $\rightarrow$ DistilBERT) remains out of scope.

7 Discussion and Limitations

Scope. The method requires white-box access and residual architecture. Non-residual networks lack the identity-target constraint that creates the fingerprint.

Weight-level, not behavioral. The signature is a property of weights, not function. Distilled students ($\mathcal{L} \approx 0.08$) are correctly classified as unrelated despite functional similarity. The method cannot detect training-data overlap or knowledge transfer.

Method	Type	Data-free	AUROC	Gap-Z ↑	Block Corr.
Diagonal Dominance (ours)	Weight-space	✓	1.000	+53.0	✓
Aligned Frobenius	Weight-space	✓	1.000	+72.0	✓
Singular Value Distance	Weight-space	✓	1.000	+ 7.9	✗
Weight Cosine	Weight-space	✓	1.000	+76.3	✗
SVCCA (Raghu et al. 2017)	Activation-space	✗	1.000	+45.4	✗
CKA (Kornblith et al. 2019)	Activation-space	✗	0.829	+ 8.6	✗
IPGuard (Cao, Jia, and Gong 2021) (regression analogue)	Decision-boundary	✗	0.500	− 3.5	✗

Table 5: Strong-baseline comparison on the MLP lineage benchmark (52 pairs). **AUROC**: overall related vs. unrelated separation; saturated for the four data-free weight-space methods. **Gap-**Z: standardized distillation rejection, $(\mu_{\text{related}} - \mu_{\text{distill}})/\sigma_{\text{distill}}$, where μ_{related} averages over the five related-checkpoint kinds; large positive values mean distilled unrelated checkpoints are cleanly below truly related ones. IPGuard scores at -3.5 because related checkpoints are ranked *below* distilled. CKA scores at $+8.6$: large in σ units only because $\sigma_{\text{distill}} = 0.014$ is tiny; the underlying ratio is $0.999/0.827 \approx 1.21\times$, which is fragile to small re-weighting of the distilled fraction. Gap-Z is scale-sensitive when σ_{distill} is very small (CKA) or when raw scores are uniformly compressed (singular-value distance, where $+7.9$ understates the gap); the decisive cross-method evidence is the raw per-kind adjacency reported in Table 9, where all four weight-space methods place distilled students at or below the diff-seed baseline while CKA and SVCCA do not. **Block Corr.**: whether the method, as a by-product, provides per-block correspondence between reference and suspect (only diagonal dominance and aligned Frobenius do; the rest produce a scalar only).

Condition	**Verdict**
Different architectures	INCOMPATIBLE
Non-residual networks	NO SIGNAL
Distillation (cross-arch)	OUT OF SCOPE
Training data overlap	NOT DETECTED
Two indep. same-arch	CANNOT DISTINGUISH

Table 6: What the method does NOT do.

Symmetric, no direction. $\mathcal{L}(A, B) = \mathcal{L}(B, A)$ by construction, so the protocol establishes *whether* two checkpoints share weight lineage but not *which* is the ancestor. Directional attribution requires either training-time metadata or comparison against a third candidate (App. C, branching tree).

Lineage validation scope. Lineage AUROC is reported on the synthetic MLP zoo and a CIFAR-10 ResNet-18 benchmark, with real-architecture evidence from block-correspondence pairing on Transformer families (Table 3) and a 7-pair real-LLM lineage test on the public LLaMA-2 family against a real cross-model null (App. D.4). The latter identifies three real fine-tuned variants—the RLHF chat model, the Vicuna instruction fine-tune, and CodeLlama—as RELATED, while four independently-trained negatives (Mistral-7B, OpenLLaMA-7B, Yi-6B, and a Llama-3-derived DeepSeek distill) all score $\mathcal{L} \approx 4 \times 10^{-4}$ and form the null, giving ≈1850× mean separation versus 6× for the strongest baseline. With only four real independents this is a descriptive separation check rather than a calibrated FPR estimate. The remaining gap is tree breadth: all related checkpoints here derive from a single base. Extending to additional real checkpoint trees (a Mistral or Qwen base with its own fine-tunes) and a larger negative set that supports a calibrated FPR is the natural next step. The per-block branch products fit comfortably in memory at 7B scale, so the gap is engineering rather than methodological.

Attacks. Function-preserving symmetries (permutations, rescaling) leave $\mathcal{L} = 1$ exactly-these are invariances, not vulnerabilities. Gradient-based suppression requires ≥12% utility loss to reach the detection threshold; full suppression destroys the model.

Positioning. This complements watermarking: watermarks require forethought; diagonal dominance offers passive, retroactive verification for checkpoints never watermarked.

8 Conclusion

Training leaves traces. Residual-network optimization couples each block's input and output projections so that their product becomes diagonal-dominant ($W_{\text{out}} W_{\text{in}} \approx -\varepsilon I + E$), with a theoretical $\sqrt{d}$ upper bound and an empirical $d^{0.87}$ growth observed from GPT-2 through LLaMA-2-13B and Mixtral-8x7B. Across 12 architectures spanning 5 families, the fingerprint recovers 91–100% of block correspondences and achieves AUROC=1.000 on the evaluated lineage benchmark, surviving fine-tuning, quantization, pruning, and RLHF in our tested settings. Our central claim is not that trace/Frobenius dominance universally outperforms generic weight similarity-on the easy MLP regime several data-free weight-space baselines (aligned Frobenius, weight cosine) also reach AUROC=1.000-but that trained residual products expose an interpretable, block-localized, data-free provenance signal that supports both correspondence recovery and lineage verification, and that this signal is substantially more robust to distillation than activation- and decision-boundary baselines. Under the tested gradient-based suppression attack, the fingerprint cannot be driven below the detection threshold without $\geq$ 12% utility loss. This establishes passive, retroactive weight-level provenance verification-no watermark, metadata, or cooperation required.

Reproducibility Statement

All results in this paper are produced by deterministic scripts with fixed seeds. The 52-pair MLP lineage benchmark (Tables 4, 5) is constructed from 2 trained reference checkpoints, each paired with 5 related-checkpoint kinds $\times 6$ derivative seeds $=$ 30 truly-related pairs (fine-tune, fine-tune to a new target, $\sigma_{\text{rel}} \in [0.01, 0.15]$ Gaussian noise, 10–85% magnitude pruning, and 16–256-level uniform quantization), plus 16 different-seed same-task pairs and 6 distilled-student pairs as unrelated checkpoints. The expanded 159-pair set used in Figure 3 draws from the full 24-checkpoint MLP zoo with the same transformation grid. Exact parameters are listed in Appendix A; null calibration uses the 50 different-seed pairs from Appendix C. Real-architecture pairing (Table 3) loads public HuggingFace checkpoints. Code, configuration files, baseline implementations (aligned Frobenius, singular-value distance, weight cosine, CKA, SVCCA, IPGuard regression analogue), the per-pair score JSONs, and the plotting scripts will be released upon acceptance.

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A Extended Experimental Details

A.1 Initialization Ablation

We test whether the diagonal-dominance fingerprint depends on initialization scheme. We train depth-24 residual MLPs (h=64, d=24) on a synthetic regression target for 200 epochs from seven initialization schemes: Kaiming uniform/normal, Xavier uniform/normal, Gaussian, uniform, and orthogonal. Table 7 reports pair accuracy before and after training.

Init scheme	Untrained	Trained	AUC
Orthogonal	0.0%	100%	0.947
Kaiming-normal	2.1%	97%	0.981
Kaiming-uniform	2.1%	98.6%	0.98
Xavier-normal	2.1%	100%	0.990
Xavier-uniform	2.1%	100%	0.99
Uniform	2.1%	93.8%	—
Gaussian σ=0.02	2.1%	13%	0.671

Table 7: Initialization ablation on depth-24 residual MLPs (3 seeds, 200 epochs). All schemes show chance-level accuracy ($\leq$2.1%) at initialization, rising to 75–100% after training. Orthogonal initialization—which provides dynamical isometry at init—shows *zero* correct pairs before training.

Critically, orthogonal initialization provides dynamical isometry at initialization by construction (Saxe, McClelland, and Ganguli 2014), yet shows *zero* correct pairs before training, ruling out the hypothesis that near-orthogonal Jacobians alone create the fingerprint. The σ=0.02 small-Gaussian case fails because residual blocks collapse to near-zero contribution; without active per-block contribution there is no Jacobian constraint to enforce.

A.2 PlainNet Control Experiment

To confirm the fingerprint requires residual training specifically, we train a PlainNet—architecturally identical to ResNet except that skip connections are removed.

While both networks converge to similar evaluation loss (1.35 vs 1.56), ResNet achieves 100% pair accuracy with 100% of correctly paired products exhibiting negative trace, while PlainNet achieves only 3%—at the chance baseline—with AUC 0.51 and no discernible diagonal structure. The skip connection creates a functional constraint that imprints the characteristic trace structure.

Architecture	Eval Loss	Pair Acc	AUC	Neg Trace
ResNet-24	1.35	100%	0.98	100%
PlainNet-24	1.56	3%	0.51	47%

Table 8: ResNet vs. PlainNet control. Despite similar evaluation loss, PlainNet achieves only chance-level pair accuracy (3% vs. chance $\approx$ 4.2%), confirming the skip connection is necessary for the fingerprint.

A.3 Training Dynamics

The fingerprint emerges rapidly during training. On a 48-block residual network:

- Epoch 0: Mean correct-pair score = 0.19, pair accuracy = 6.25%
- Epoch 5: Diagonal fully separated, pair accuracy = 100%
- Epoch 300: Mean correct-pair score = 2.07, mean incorrect-pair score = 0.16 (21$\times$ ratio)

The transformation occurs within the first few epochs, well before training loss plateaus.

B Extended Theoretical Results

B.1 Null Model (Corollary)

Corollary 2 (Null Model)**.** *For randomly initialized weights with no training, all pairs satisfy* $\mathbb{E}[s(i,j)] = 1/\sqrt{d} + o(1)$ *uniformly in* (i,j)*. Hungarian matching recovers correct pairs at the chance rate* $1/L$*.*

The null model rules out three alternative explanations:

1. Diagonal dominance is not an artifact of compatible matrix shapes
2. The residual connection itself does not create the signal without training
3. Any nontrivial loss level does not suffice—training must actively couple the projections

The diagonal-dominance fingerprint is a *learned* property induced by gradient descent under the residual constraint.

B.2 Geometric Interpretation of the Score

The diagonal-dominance score $s(M) = |\mathrm{tr}(M)|/\|M\|_F$ has a natural geometric interpretation:

- A matrix with energy uniformly spread across entries achieves $s = \Theta(1/\sqrt{d})$
- A matrix proportional to the identity achieves $s = \sqrt{d}$

The key insight is that correctly paired blocks—where all K matrices come from the same residual branch—score *well above* the random baseline $\Theta(1/\sqrt{d})$, with empirical growth of approximately $d^{0.87}$ across $d \in [768, 5120]$ (Appendix D), bounded above by $\sqrt{d}$ but typically reaching only 15–25% of that bound. The signal-to-baseline ratio therefore grows roughly as $d^{1.37}$, making correct pairs increasingly separable in wider networks.

For typical trained blocks, $\|E\|_F/(\varepsilon\sqrt{d}) \in [1.9, 3.8]$, yielding $s(i,i) \in [1.76, 3.23]$—well above the baseline but below the theoretical maximum of $\sqrt{d}$.

C Verification Protocol Details

C.1 Gated Branch Score

Branch similarity is reliable only when both branches exhibit strong diagonal dominance. We gate the cosine similarity:

$$G(\phi_\ell^A, \phi_m^B) = \langle \phi_\ell^A, \phi_m^B \rangle \cdot \min\left(\frac{s(M_\ell^A)}{\tau_s}, \frac{s(M_m^B)}{\tau_s}, 1\right) \quad (10)$$

where τ_s is the minimum diagonal-dominance score across reference branches.

C.2 Statistical Calibration

We calibrate against a null distribution $\mathcal{N}_A = \{\mathcal{L}(A, U_j) : U_j \notin \mathcal{D}(A)\}$ of scores between reference A and independently trained models. The z-score is:

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_{\text{null}}}{\sigma_{\text{null}}} \quad (11)$$

Under Gaussian approximation, $\tau = 1.645$ yields 95% confidence; $\tau = 3.0$ yields FPR $< 0.13\%$ for high-stakes applications.

C.3 Full Protocol Steps

1. **Compatibility Check:** Verify $\text{arch}(A) = \text{arch}(B)$. If not, return INCOMPATIBLE.
2. **Extract:** Compute architecture-aware branch products $\{M_\ell^A\}$, $\{M_\ell^B\}$.
3. **Fingerprint:** Compute centered residual signatures $\{\phi_\ell^A\}$, $\{\phi_\ell^B\}$.
4. **Align:** Apply Hungarian matching to recover block correspondences.
5. **Aggregate:** Compute lineage score $\mathcal{L}(A, B)$.
6. **Calibrate:** Compute z-score $Z(A, B)$.
7. **Decide:** Return verdict:

$$\text{Verdict} = \begin{cases} \text{RELATED} & \text{if } Z(A,B) > \tau_{\text{upper}} \\ \text{UNRELATED} & \text{if } Z(A,B) < \tau_{\text{lower}} \\ \text{INCONCLUSIVE} & \text{otherwise} \end{cases} \quad (12)$$

C.4 Failure Modes

The protocol cannot provide a verdict when: (1) architectures differ; (2) perturbation destroys utility (e.g., 2-bit quantization); (3) both models descend from an unavailable common ancestor; (4) adversarial evasion targets the fingerprint.

C.5 Lineage Chain and Branching Tree

To test multi-generational ancestry recovery, we construct a branching tree: root C_1, siblings C_2, C_3 fine-tuned from C_1, grandchildren $C_4 = C_2 \to$ FT, $C_5 = C_3 \to$ FT. The lineage score recovers correct ancestry: for C_4, ranking by $\mathcal{L}(C_4, \cdot)$ is parent (0.96) $>$ grandparent (0.91) $>$ uncle (0.87) $>$ cousin (0.85) $>$ independent (0.08). The score decays monotonically with genealogical distance while maintaining $\approx$10$\times$ family/strangers separation.

D Per-Architecture Details

D.1 GPT-2 Scaling

Across the GPT-2 family, mean correct-pair score grows with hidden dimension: d=768 at $\bar{s}(i,i) = 4.18$, d=1024 at 5.46, d=1280 at 6.98, d=1600 at 7.77. The fitted exponent within the GPT-2 family is ≈ 0.85, consistent with the cross-family $s \propto d^{0.87}$ scaling reported in the main text and exceeding the $\sqrt{d}$ envelope rate; the $\sqrt{d}$ upper bound itself is never violated (observed scores reach 15–19% of the bound).

D.2 Alternative Methods Comparison

Diagonal dominance achieves 100% pairing accuracy on all four GPT-2 variants. Frobenius norm matching degrades from 67% (GPT-2) to 47% (GPT-2-large). Singular-value distance performs at chance (0–8%). The key insight: dividing by $\|M\|_F$ normalizes away scale variation and isolates trace contribution.

D.3 Per-Kind Lineage Baseline Breakdown

Table 9 reports the mean lineage score of each baseline broken down by related-checkpoint kind (fine-tune, fine-tune to a new target, noise, prune, quantize) and unrelated-checkpoint kind (different-seed same-task, distilled). The related rows should be high; the unrelated rows should be low. The aggregate AUROC reported in Table 5 hides two structural failures that this breakdown exposes:

- **CKA and SVCCA fail on distilled unrelated checkpoints.** CKA scores distilled students at 0.83 and SVCCA at 0.89—only 1.2$\times$ and 1.13$\times$ below the corresponding fine-tuned related checkpoints (0.999 and 0.9997). On the present benchmark there are 16 different-seed unrelated checkpoints and only 6 distilled, so AUROC remains high; any reweighting toward distillation would collapse it.
- **IPGuard scores noise-perturbed related checkpoints at 0.000.** The decision-boundary analogue ranks all 6 noise-perturbed related checkpoints below all unrelated ones, driving overall AUROC to chance (0.500).

All four weight-space methods (diagonal dominance, aligned Frobenius, singular-value distance, weight cosine) score distilled unrelated checkpoints at or below the different-seed baseline—they cleanly reject distillation, while the activation- and decision-boundary methods cannot.

D.4 Real-LLM Lineage Validation

To move beyond block-correspondence pairing on real foundation models, we evaluate $\mathcal{L}(\cdot, \cdot)$ on the public LLaMA-2 family against the weight-space and decision-boundary baselines of Section 5.3. Using LLaMA-2-7B as reference, we score seven real reference-vs-model pairs: three real fine-tuned *related* checkpoints and four independently-trained *unrelated* checkpoints. For each model we extract the per-layer SwiGLU branch product $M_\ell = W_{\text{down}}^{(\ell)} W_{\text{up}}^{(\ell)} \in \mathbb{R}^{d \times d}$, form the centered residual signature ϕ_ℓ, and score

Method	Related (should be high)					Unrelated (should be low)	
	FT	FT-new tgt	Noise	Prune	Quant	Diff-seed	Distilled
Diag. Dominance (ours)	0.391	0.326	0.400	0.359	0.404	0.029	**0.030**
Aligned Frobenius	−0.036	−0.301	−0.026	−0.204	−0.007	−1.333	−1.300
Singular Value Dist.	−0.004	−0.023	−0.003	−0.018	−0.001	−0.030	−0.032
Weight Cosine	0.999	0.956	1.000	0.979	1.000	0.096	0.109
SVCCA	1.000	0.953	1.000	0.993	1.000	0.881	*0.885*
CKA	0.999	0.760	1.000	0.986	1.000	0.820	*0.827*
IPGuard (regr.)	0.888	0.110	*0.000*	0.626	0.993	0.632	0.642

Table 9: Per-kind mean lineage scores on the 52-pair MLP benchmark. Each method uses its native scale; what matters is the relative gap between the rightmost two columns (unrelated) and the leftmost five (related). Italicized entries highlight failure modes: SVCCA and CKA give distilled unrelated checkpoints near-related scores; IPGuard scores noise-perturbed related checkpoints at exactly 0.000. All four weight-space methods (including ours) keep distilled below or at the diff-seed baseline. Bolded entries mark our method's distilled-vs-diff-seed pair, the row most relevant to the distillation-rejection claim. n=6 per related kind, n=16 diff-seed, n=6 distilled.

$\mathcal{L}(A, B) = \frac{1}{L}\sum_\ell \langle \phi_\ell^A, \phi_\ell^B \rangle$. The same branch products feed the three weight-space baselines (weight cosine, aligned Frobenius, singular-value distance); IPGuard is computed separately from output logits and is available only on the subset of pairs for which logits were collected (2 related, 2 unrelated), and the activation baselines (CKA, SVCCA) are omitted at 7B scale because per-pair computation over 32×32 layers is prohibitively expensive. The benchmark runs on a single GPU via `gpu_benchmark_issue44.py`.

Compatibility. The strict protocol (App. C) returns INCOMPATIBLE when $\mathrm{arch}(A) \neq \mathrm{arch}(B)$. Because M_ℓ contracts the intermediate width d_{ff} and lives in $\mathbb{R}^{d\times d}$, the branch-product comparison itself needs only matching depth L and hidden size d. For the two cross-family negatives (Mistral and DeepSeek carry $d_{\mathrm{ff}} = 14336$ vs. 11008 for the LLaMA/Yi models) we therefore apply this *relaxed* branch-product compatibility rather than strict architecture equality; we flag it as a deliberate relaxation used only to admit cross-family negatives. Note that OpenLLaMA—the decisive negative—satisfies the strict rule exactly, so the strongest negative does not rely on the relaxation.

Pairs. Related: the official RLHF model Llama-2-7B-chat, the Vicuna-7B instruction fine-tune, and CodeLlama-7B, a code-specialized continued-pretraining derivative of the base. *Unrelated:* four independently-trained checkpoints with no LLaMA-2 weight ancestry—Mistral-7B-v0.1, OpenLLaMA-7B, Yi-6B, and DeepSeek-R1-Distill-Llama-8B. The last is *behaviorally* distilled from DeepSeek-R1 reasoning traces but *weight-derived* from Meta's Llama-3.1-8B; its UNRELATED verdict w.r.t. LLaMA-2 demonstrates that the method tracks weight inheritance rather than behavioral pedigree, consistent with Section 5.3.

OpenLLaMA: a same-architecture, from-scratch negative. OpenLLaMA-7B is an open reproduction of the LLaMA architecture trained from scratch on different data. It shares LLaMA-2's exact residual shape ($d = 4096$, $L = 32$, $d_{\mathrm{ff}} = 11008$), tokenizer conventions, and RoPE setup, yet inherits none of its weights. It is therefore the strongest possible negative: architecture is held *constant* and only weight lineage varies, so a configuration-level check cannot distinguish it from a genuinely related checkpoint—the verdict must come from the weights. Diagonal dominance scores it at $\mathcal{L} = 4.5 \times 10^{-4}$, indistinguishable from the other unrelated checkpoints.

Cross-model null. The four unrelated checkpoints define a real cross-model null $\mu_0 = 4.2 \times 10^{-4}$, $\sigma_0 = 5.0 \times 10^{-5}$ (all four within $[3.5, 4.8] \times 10^{-4}$). With only $n = 4$ real independents this is a *descriptive* separation check rather than a calibrated false-positive-rate estimate; we therefore report the raw $\mathcal{L}$ and the separation in orders of magnitude as the primary evidence, and defer a calibrated FPR to the larger checkpoint trees discussed below. The four unrelated checkpoints span three from-scratch pre-training lineages (Mistral, OpenLLaMA, Yi) plus a Llama-3 derivative, so the null is not dominated by any single family. *Verdict rule for Table 10:* we label a pair RELATED if $\mathcal{L}$ exceeds a descriptive threshold placed an order of magnitude above the null ($\mathcal{L} > 10^{-2}$) and UNRELATED otherwise; all seven pairs fall unambiguously on the expected side, with the nearest related checkpoint (CodeLlama, $\mathcal{L} = 0.336$) still more than two orders of magnitude above the null.

Baseline separation. Aggregated over the seven pairs, diagonal dominance separates related from unrelated checkpoints by ≈1850× in mean score (0.78 vs. 4.2×10^{-4}), versus 6× for weight cosine (0.80 vs. 0.13); IPGuard, available only on the four-pair logit subset, gives 2× (0.25 vs. 0.13); aligned Frobenius and singular-value distance do not separate this real-model set (AUROC 0.667 each), as their global spectral/Frobenius statistics are dominated by cross-model scale differences. This reproduces on real 7B weights the margin advantage established on the synthetic and CIFAR benchmarks: several methods clear AUROC on easy splits, but the residual-signature score preserves the largest related-vs-unrelated margin.

The set already spans a real RLHF model, a real instruction fine-tune, and a real continued-pretraining derivative (CodeLlama), tested against the strongest available

Pair (base vs.)	Kind	$\mathcal{L}$	W. cos	Verdict
Related (real fine-tunes)				
Llama-2-7B-chat	RLHF	0.995	0.995	REL.
Vicuna-7B	instruct	0.996	0.996	REL.
CodeLlama-7B	code PT	0.336	0.414	REL.
Unrelated (cross-model null)				
Mistral-7B-v0.1	scratch	$4e{-4}$	0.158	UNR.
OpenLLaMA-7B	scratch†	$4e{-4}$	0.089	UNR.
Yi-6B	scratch	$5e{-4}$	0.122	UNR.
DeepSeek-Distill-8B	L3.1 wts	$4e{-4}$	0.162	UNR.

Table 10: Real-LLM lineage validation on the public LLaMA-2 family (7 real pairs). All seven verdicts match expectation; REL. / UNR. abbreviate RELATED / UNRELATED. †OpenLLaMA shares LLaMA-2's exact architecture (d=4096, L=32, d_{ff}=11008) but is trained from scratch, so its classification as unrelated cannot come from configuration. Diagonal dominance ($\mathcal{L}$) drives every unrelated checkpoint to the 10^{-4} null while keeping related checkpoints at ≥ 0.34; weight cosine (W. cos) also separates but compresses the unrelated only to ≈0.13. CodeLlama, a heavy code-specialized continued-pretraining fine-tune, is the weakest related checkpoint ($\mathcal{L} = 0.336$) yet still ∼700× above the null—graceful degradation with fine-tune intensity rather than a sharp threshold. All models share $d = 4096$, $L = 32$; branch products are $d \times d$ regardless of d_{ff}.

negative—a same-architecture from-scratch model (OpenLLaMA) whose classification as unrelated cannot be explained by configuration alone. The remaining limitation is tree breadth: all related checkpoints here derive from a single base (LLaMA-2-7B). Extending to additional real checkpoint trees (e.g. a Mistral or Qwen base with its own community fine-tunes) and to a larger negative set that supports a calibrated FPR is the natural next step.

D.5 Harder Regime: Layer Grafts and Linear Merges

To probe the regime predicted by Section 5.3 where a flat global cosine should blur partial provenance, we evaluate all seven baselines on two harder constructions over the depth-16 MLP zoo: (i) **layer grafts**, where a suspect is built by copying $K \in \{0, 4, 8, 12, 16\}$ of the reference's 16 blocks into a same-task independent donor (and labelling $K \geq 8$ as related); and (ii) **linear merges**, where suspect parameters equal $w \cdot \text{ref} + (1 - w) \cdot \text{donor}$ with $w \in \{0, 0.25, 0.5, 0.75, 1.0\}$ (related if $w \geq 0.5$). We use 2 trained references, 2 trained donors per reference, and report AUROC for binary related-vs-unrelated detection plus Spearman correlation between each method's score and the reference fraction (K/L or w); the latter measures whether a method tracks partial overlap monotonically rather than just clearing a threshold. Table 11 summarizes the 40-pair benchmark.

	AUROC		Spearman ρ	
Method	Graft	Merge	Graft	Merge
Diagonal dominance (ours)	0.979	1.000	+0.96	+0.96
Aligned Frobenius	1.000	1.000	+0.99	+0.99
Weight cosine	1.000	1.000	+0.98	+0.96
Singular-value distance	1.000	0.448	+0.99	+0.23
SVCCA	1.000	1.000	+0.97	+0.99
CKA	0.813	0.781	+0.73	+0.75
IPGuard (regr. analogue)	0.052	0.448	−0.83	+0.22

Table 11: Harder-regime benchmark on layer grafts and linear merges (40 pairs total). Related is defined as $K \geq L/2$ for grafts and $w \geq 0.5$ for merges. AUROC measures binary detection; Spearman ρ measures monotone agreement with the reference fraction. Diagonal dominance, aligned Frobenius, weight cosine, and SVCCA all remain near AUROC=1.000 with $\rho \geq 0.96$ on both regimes—confirming Section 5.3's framing that the contribution is *not* that residual-product dominance beats generic weight cosine on aggregate detection. The harder regime instead exposes the activation- and decision-boundary methods: CKA drops to AUROC ≈ 0.80 with $\rho \approx 0.74$, IPGuard collapses entirely (AUROC 0.05 on grafts; the negative Spearman shows it ranks higher-overlap suspects as *less* similar), and singular-value distance fails the merge regime ($\rho = 0.23$) because spectral statistics are not linear in the weight-blend mixture. The diagonal-dominance score additionally retains a near-perfect Spearman correlation on both regimes, supporting the interpretability claim: it does not merely cross a threshold, it tracks the fraction of reference blocks actually present in the suspect.

D.6 ResNet Factorization

For Bottleneck blocks, the naive W_3W_1 achieves chance-level accuracy. The triple product $W_3W_2W_1$ recovers:

- ResNet-50/layer3 (n=5): 100%, AUC 1.000
- ResNet-101/layer3 (n=22): 100%, AUC 0.995
- ResNet-152/layer3 (n=35): 91%, AUC 0.964

D.7 CIFAR-10 ResNet-18 Benchmark

We train two CIFAR-10 ResNet-18 references and three independents. Each reference yields 11 related checkpoints (noise 1−10%, pruning 20−80%, quantization 16−256 levels). Results: AUROC=1.000, TPR@1%FPR=100%. Related checkpoint scores: $\mathcal{L} \in [0.695, 1.000]$; unrelated baseline: $\mathcal{L}_{\max} = 0.004$.

D.8 Branching Ancestry Recovery

On a synthetic tree $C_1 \to \{C_2, C_3\}$, $C_2 \to C_4$, $C_3 \to C_5$, the lineage score ranks candidates correctly:

$$\begin{aligned}\mathcal{L}(C_4, C_2) = 0.961 \;>\; \mathcal{L}(C_4, C_1) = 0.910 \\ >\; \mathcal{L}(C_4, C_3) = 0.867 \;>\; \mathcal{L}(C_4, C_5) = 0.850 \\ \gg\; \mathcal{L}(C_4, I_k) \approx 0.08.\end{aligned}$$

The score decays monotonically with genealogical distance while maintaining ≈10× family/strangers separation.

E Attack and Robustness Details

E.1 Gradient-Based Suppression Attack

An adversary optimizes perturbation Δ to minimize lineage score subject to utility:

$$\min_{\Delta} \mathcal{L}_{\cos}(A, A+\Delta) \; + \; \lambda \cdot \|f_{A+\Delta}(X) - f_A(X)\|_2^2 \quad (13)$$

λ	Final $\mathcal{L}$	Eval loss	Verdict
0	0.053	39.5	utility destroyed
10^{-2}	0.054	0.77	+12% loss
10^{-1}	0.083	0.70	+1.5% loss
≥ 1	0.91+	0.69	utility preserved

Table 12: Pareto frontier: suppressing $\mathcal{L}$ to chance level requires $\geq$12% utility loss.

E.2 Reparameterization Invariances

Permutations: $W_{\text{in}} \leftarrow PW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}P^\top$ leaves $M = W_{\text{out}}W_{\text{in}}$ unchanged.

Rescaling: $W_{\text{in}} \leftarrow cW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}/c$ preserves M exactly.

Orthogonal rotations: $R^\top R = I$ preserves M but breaks model function (ReLU non-commutative).

All three yield $\mathcal{L} = 1.0$ to numerical precision—these are invariances, not evasion paths.

F Practical Applications

The fingerprint enables applications beyond forensic verification.

F.1 Training Quality Assurance

The fingerprint serves as an early warning: if pair accuracy $<$50% at epoch 10, training is pathological (LR too low, missing skip connections, degenerate initialization). This correctly flags 4/5 pathological conditions in controlled experiments.

F.2 Zero-Knowledge Ownership Proofs

The E matrix ($M = -\varepsilon I + E$) provides a commitment scheme: owners commit to Hash$(E_k \| r_k)$ at registration, reveal challenged blocks on request. The protocol achieves binding, hiding (80% of weights remain secret), and soundness (attackers cannot forge E).

F.3 Compression Auditing

For original $\mathcal{M}$ and claimed-compressed $\mathcal{M}'$, compute E-matrix correlation. Quantization ($r > 0.97$), pruning ($r > 0.75$ at 70%), and fine-tuning ($r > 0.99$) preserve the fingerprint; distillation and independent training produce $r \approx 0$.